ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** biotic components	Living things.
2	**(b)** food chain	Linear.
3	**(c)** ultraviolet rays	UV.
4	**(d)** Sunlight	Abiotic.
5	**(b)** herbivores	Primary consumers.
6	**(d)** three atoms of oxygen	O3.
7	**(a)** producers	They make their own food.
8	**(b)** Rabbit	A herbivore.
9	**(d)** CFCs	Chlorofluorocarbons.
10	**(c)** herbivores	Plant-eaters.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Living and non-living components interacting with each other (2).	2
12	Bacteria (1); Fungi (1).	1+1
13	Chlorofluorocarbons (2).	2
14	Temperature (1); Rainfall, soil or light (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Natural: forest, pond, lake (1); Artificial: aquarium, garden, crop field (1); Natural ecosystems are not made by humans (1).	1+1+1
16	10% passes on (1); 10 per cent of 5000 (1); 500 J (1).	1+1+1
17	Biodegradable: vegetable peels, paper, cotton (1); Non-biodegradable: plastic, DDT, glass (1); Microbes act on biodegradable ones (1).	1+1+1
	OR: Producers: grass (1); Consumers: deer, tiger (1); Decomposers: mushroom, bacteria (1).	1+1+1
18	Second level: 100 J (1); Third level: 10% of 100 (1); 10 J (1).	1+1+1
	OR: It is non-biodegradable (1); It pollutes soil and water (1); Animals eat it and may die (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Ecosystem: living and non-living parts (1); Abiotic: air, water, sunlight (1); Biotic: producers (1); Biotic: consumers (1); Biotic: decomposers (1).	1+1+1+1+1
	OR: Harmful pesticide enters the water (1); Taken by plants (1); Eaten by herbivores (1); Eaten by carnivores (1); Highest concentration at the top level (1).	1+1+1+1+1
20	Producers make food (1); Consumers eat food (1); Herbivores, carnivores and omnivores (1); Decomposers break down remains (1); Nutrients are returned to the soil (1).	1+1+1+1+1
	OR: Biodegradable: decomposed by microbes (1); Examples: food waste, paper (1); Non-biodegradable: not decomposed (1); Examples: plastics, DDT (1); They cause pollution (1).	1+1+1+1+1

Mistakes to avoid:

- Drawing arrows in a food chain from the eater to the food; the arrow points from the organism that is eaten to the one that eats it.
- Writing that 90% of the energy passes on to the next level; only 10% is transferred and 90% is lost, so energy flows in one direction and does not cycle.
- Counting decomposers as a trophic level of the food chain, or calling the Sun a producer; green plants are producers, the Sun is the source of energy.
- Confusing ozone (O_3) with oxygen (O_2), and writing that CO_2 (instead of CFCs) is the main cause of ozone depletion.

- Saying that all waste is non-biodegradable or that plastics decompose quickly; plastics, glass and DDT are non-biodegradable.
- Explaining biomagnification as increase of pesticide in producers only; the concentration is highest in the **top** consumers.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Sun	Producers use it.
2	**(b)** non-biodegradable	They stay for years.
3	**(d)** carnivores	Meat-eaters.
4	**(b)** producer	Plants.
5	**(b)** Leaves	Microbes act on them.
6	**(a)** omnivores	Like humans and crows.
7	**(d)** food web	Network.
8	**(b)** DDT	A persistent pesticide.
9	**(b)** carnivore	Eats animals.
10	**(b)** 10%	10 per cent law.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A step in a food chain (2).	2
12	Stratosphere (2).	2
13	An organism that makes its own food (1); By photosynthesis (1).	1+1
14	Grass, deer, tiger (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Herbivore: eats plants (goat) (1); Carnivore: eats flesh (lion) (1); Omnivore: eats both (humans) (1).	1+1+1
16	A chain is a single path (1); A web has many paths (1); A web is more stable (1).	1+1+1
17	Reduce use of things (1); Reuse them (1); Recycle them (1).	1+1+1
	OR: The Sun (1); Producers capture sunlight (1); Consumers eat producers and get energy (1).	1+1+1
18	Energy is used for life processes (1); Some is lost as heat (1); Only 10% passes to the next level (1).	1+1+1
	OR: Vegetable waste is converted into manure (1); Reduces garbage (1); Reduces pollution (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Series of organisms feeding on each other (1); Grass, grasshopper, frog, snake, hawk (1); Grass: producer (1); Grasshopper: primary consumer; frog: secondary (1); Snake and hawk: higher consumers (1).	1+1+1+1+1
	OR: Segregate biodegradable and non-biodegradable waste (1); Compost the biodegradable waste (1); Recycle paper and glass (1); Reduce plastic use (1); Reuse bags and bottles (1).	1+1+1+1+1
20	Only 10% energy is passed on (1); Producers: 10,000 J (1); Herbivores: 1000 J (1); Carnivores: 100 J (1); Top carnivores: 10 J (1).	1+1+1+1+1
	OR: More UV reaches the earth (1); Causes skin cancer (1); Cataract in eyes (1); Reduce CFC use (1); Use eco-friendly refrigerants (1).	1+1+1+1+1

Mistakes to avoid:

- Drawing arrows in a food chain from the eater to the food; the arrow points from the organism that is eaten to the one that eats it.
- Writing that 90% of the energy passes on to the next level; only 10% is transferred and 90% is lost, so energy flows in one direction and does not cycle.
- Counting decomposers as a trophic level of the food chain, or calling the Sun a producer; green plants are producers, the Sun is the source of energy.
- Confusing ozone (O_3) with oxygen (O_2), and writing that CO_2 (instead of CFCs) is the main cause of ozone depletion.
- Saying that all waste is non-biodegradable or that plastics decompose quickly; plastics, glass and DDT are non-biodegradable.
- Explaining biomagnification as increase of pesticide in producers only; the concentration is highest in the **top** consumers.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** reduce, reuse, recycle	Environment friendly.
2	**(b)** herbivore	Eats grass.
3	**(a)** phytoplankton	They carry out photosynthesis.
4	**(b)** composting	Decomposition.
5	**(d)** decomposers	Recyclers.
6	**(c)** herbivores	Primary consumers.
7	**(d)** separate biodegradable and non-biodegradable waste	Segregation.
8	**(c)** a natural ecosystem	Not man-made.
9	**(b)** A food chain	Unidirectional.
10	**(b)** top consumers	Biomagnification.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Substance broken down by living organisms like bacteria (2).	2
12	An organism that depends on others for food (1); Cow, lion (1).	1+1
13	Only 10% of energy passes to the next trophic level (2).	2
14	Increase of harmful chemicals along the food chain (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Plant, insect, frog, snake (1); T1 producer, T2 primary consumer (1); T3 secondary consumer, T4 tertiary consumer (1).	1+1+1
16	Absorbs UV rays (1); Protects living organisms (1); Prevents skin cancer (1).	1+1+1
17	Harmful chemicals enter the food chain (1); They do not break down (1); Concentration increases at higher levels (1).	1+1+1
	OR: Break down dead organisms (1); Return nutrients to the soil (1); Keep the environment clean (1).	1+1+1
18	By CFCs (1); Used in refrigerators and aerosols (1); Chlorine destroys ozone (1).	1+1+1
	OR: Less energy is available at higher levels (1); Few organisms can be supported (1); Top consumers are very few (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Producers get energy from the Sun (1); They give it to the herbivores (1); Herbivores give to carnivores (1); Energy does not return (1); It is lost as heat (1).	1+1+1+1+1
	OR: A network of food chains (1); Algae eaten by tadpoles and small fish (1); Small fish eaten by large fish (1); Large fish eaten by birds (1); Many connections form the web (1).	1+1+1+1+1
20	A layer of ozone gas in the stratosphere (1); Absorbs UV rays (1); Protects living beings (1); Damaged by CFCs (1); Produces an ozone hole (1).	1+1+1+1+1
	OR: Natural: not made by humans (1); Artificial: man-made (1); Natural: forest, pond (1); Artificial: aquarium, garden (1); Artificial needs maintenance (1).	1+1+1+1+1

Mistakes to avoid:

- Drawing arrows in a food chain from the eater to the food; the arrow points from the organism that is eaten to the one that eats it.
- Writing that 90% of the energy passes on to the next level; only 10% is transferred and 90% is lost, so energy flows in one direction and does not cycle.
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